7.5: The content of a 4-bit shift register is initially 1101. The register is shifted six times to the right, with the serial input being 101101. What is the content of the register after each shift?

Answer:

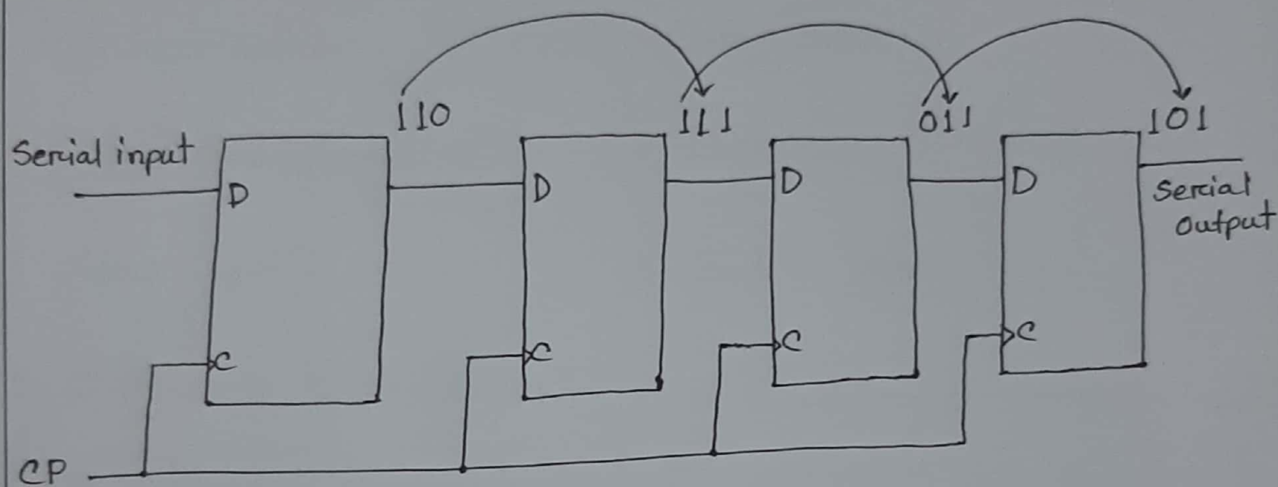


Fig: 4-bit shift register

1 0 1 1 0 1

1101 → 1110 → 0111 → 1011 → 1101 → 0110 → 1011

After six clock cycles, the content of the register will be 1110; 0111; 1011; 1101; 0110; 1011.

7.6: What is the difference between serial and parallel transfer? What type of register is used in each case?

Answer: The difference between serial & parallel transfer is given below:

Serial Transfer	Parallel Transfer
1. Data is transferred 1 bit at a time.	1. Data is transferred n-bits at a time
2. Used for long distance communication.	2. Used for short distance communication
3. Economical (simpler)	3. Expensive (complex)
4. Slow speed	4. Faster speed
5. Only one communication channel is required.	5. N number of communication channels are required for N parallel bit transfer.

Serial to Parallel transfer:

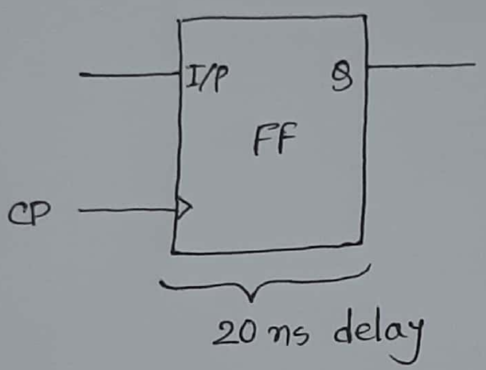
A shift register can convert serial data into parallel data by first shifting 1 bit at a time into the register and then taking the parallel data from register output.

Parallel to Serial transfer:

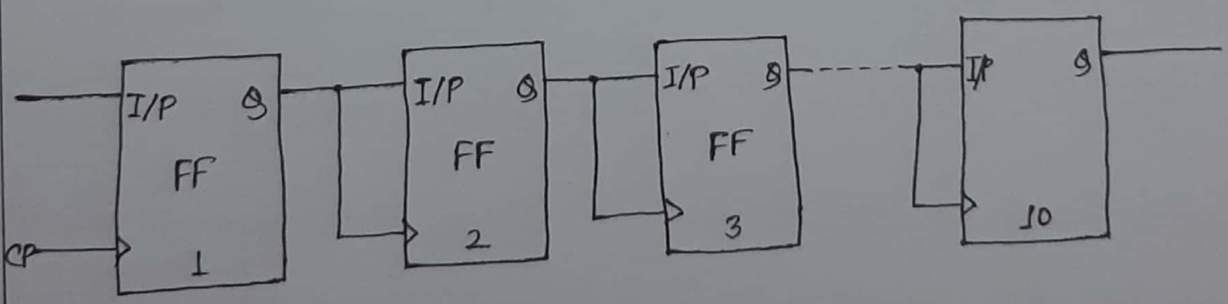
A shift register with parallel load can convert parallel data to a serial format by first loading the data in parallel, then shifting the bits one at a time.

7.11: A flipflop has a 20-ns delay from the time its CP input goes from 1 to 0. to the tune the output is complemented. What is the maximum delay in a 10-bit binary ripple counter that uses these flip-flops? What is the maximum frequency the counter can operate at reliably?

Answer:



For 10-bit binary ripple counter, there will be ten flipflops.



After 20 ns the output of the first FF will be applied to the 2nd FF and this will continue to the 10th FF. So, after 200 ns we will get the output of the 10th FF.

The worst case is when the all the FF will be complemented.

There force, the maximum delay $= 10 \times 20$

$= 200$ ns

$\therefore$ The maximum frequency $= \dfrac{1}{\text{maximum delay}}$

$$= \frac{1}{200 \times 10^{-9}} \text{ Hz}$$

$$= \frac{1}{2 \times 10^{-7}} \text{ Hz}$$

$$= 2 \times 10^{7} \text{ Hz}$$

$$= 20 \times 10^{6} \text{ Hz}$$

$$= 20 \text{ MHz}.$$

7.12: How many flipflops must be complemented in a 10 bit binary ripple counter to reach the next count after 0111111111?

Answer: First we have to add 1 to the given binary number, then we have to see how many bits of the given number are complemented after the addition. The number of bits complemented are the number of the flip-flops ~~mus~~ that must be complemented.

$$
\begin{array}{r}
0111111111 \\
+1 \\
\hline
1000000000
\end{array}
$$

Here, we can see all the 10 bits of the given number is complemented after the addition. So, here 10 flip-flops are complemented.

7.13: Draw the diagram of a 4-bit binary ripple down-counter using flip-flops that trigger on the (a) positive-edge transition and (b) negative-edge transition.

Answer:

a) Positive-edge transition-

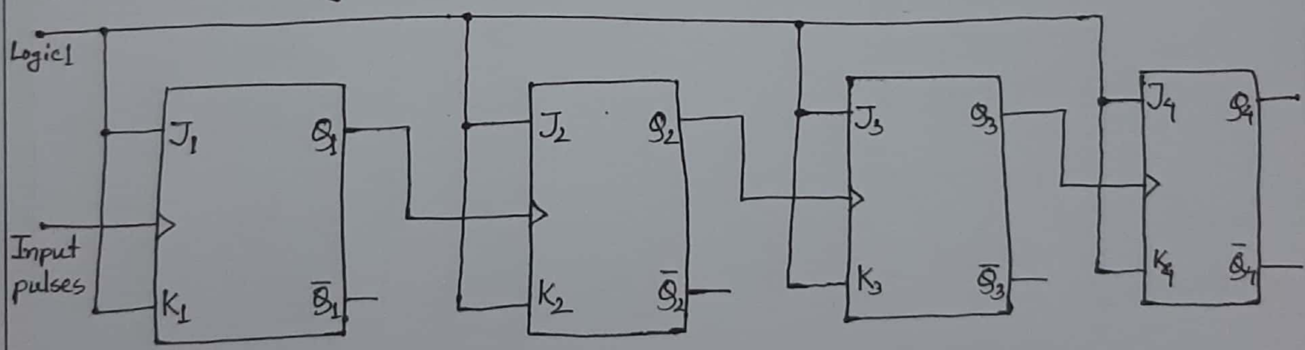


Fig: Binary ripple countdown counter

b) For negative-edge transition the diagram of a 4-bit binary ripple down counter using flip-flops ~~are~~ is given below:

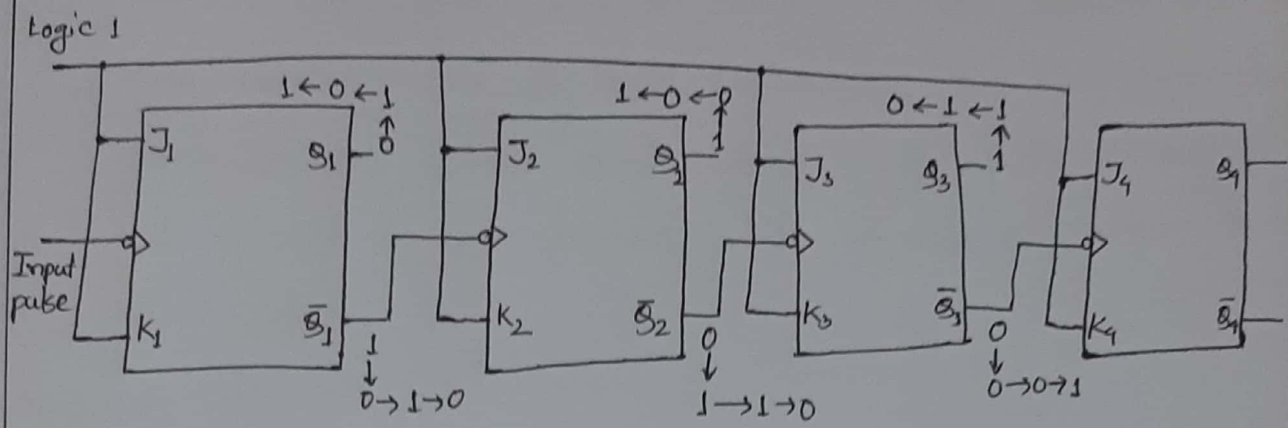
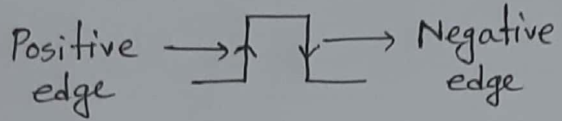


Fig: Binary ripple countdown counter

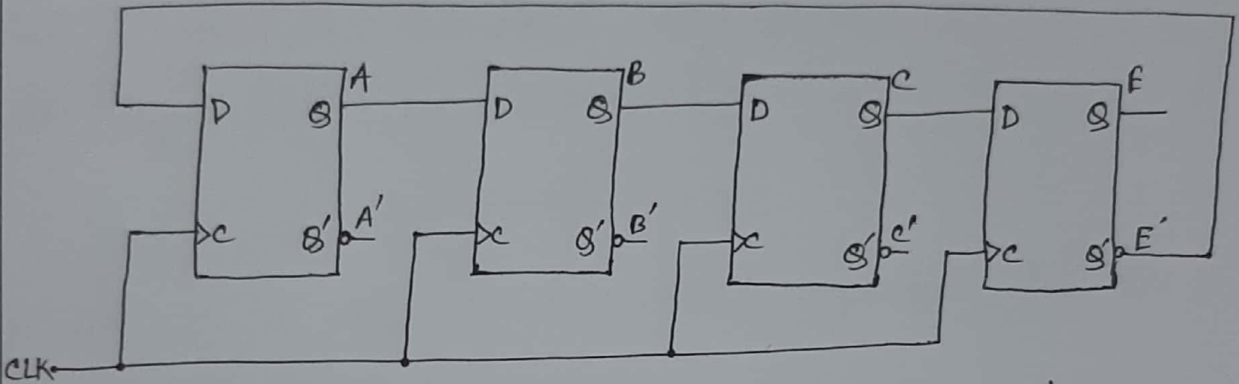


Clock	Q_4	Q_3	Q_2	Q_1	
0	0	0	0	0	
1	1	1	1	1	→ 15
2	1	1	1	0	→ 14
3	1	1	0	1	→ 13
4	1	1	0	0	→ 12
5	1	0	1	1	→ 11

down-counting

7.27: a) List the eight unused states in the switch-tail ring counter of Fig 7.23. Determine the next state for each unused state and show that, if the circuit finds itself in an invalid state, it does not return to a valid state.

Answer:



(a) 4-stage switch-tail ring counter

The valid used states sequence is : 0, 8, 12, 14, 15, 7, 3, 1

∴ The unused states are : 2, 4, 5, 6, 9, 10, 11, 13

	Present State				Next State				Invalid States
	A	B	C	E	A(t+1)	B(t+1)	C(t+1)	E(t+1)	
2 →	0	0	1	0	1	0	0	1	→ 9
4 →	0	1	0	0	1	0	1	0	→ 10
5 →	0	1	0	1	0	0	1	0	→ 2
6 →	0	1	1	0	1	0	1	1	→ 11
9 →	1	0	0	1	0	1	0	0	→ 4
10 →	1	0	1	0	1	1	0	1	→ 13
11 →	1	0	1	1	0	1	0	1	→ 5
13 →	1	1	0	1	0	1	1	0	→ 6

b) Modify the circuit as recommended in the text and show that (1) the circuit produces the same sequence of states as listed in Fig 7.23(b), and (2) the circuit reaches a valid state from any one of the unused states.

Answer: The correcting procedure recommended in the text is to disconnect the output from flip-flop B that goes to the D input of flip-flop C and instead enable the input of flip-flop C by the function

$$D_C = \cancel{(A+B)}\ (A+C)B$$

where D_C is the flip-flop input equation for the D input of flip-flop C.

So the modified circuit is as follows:

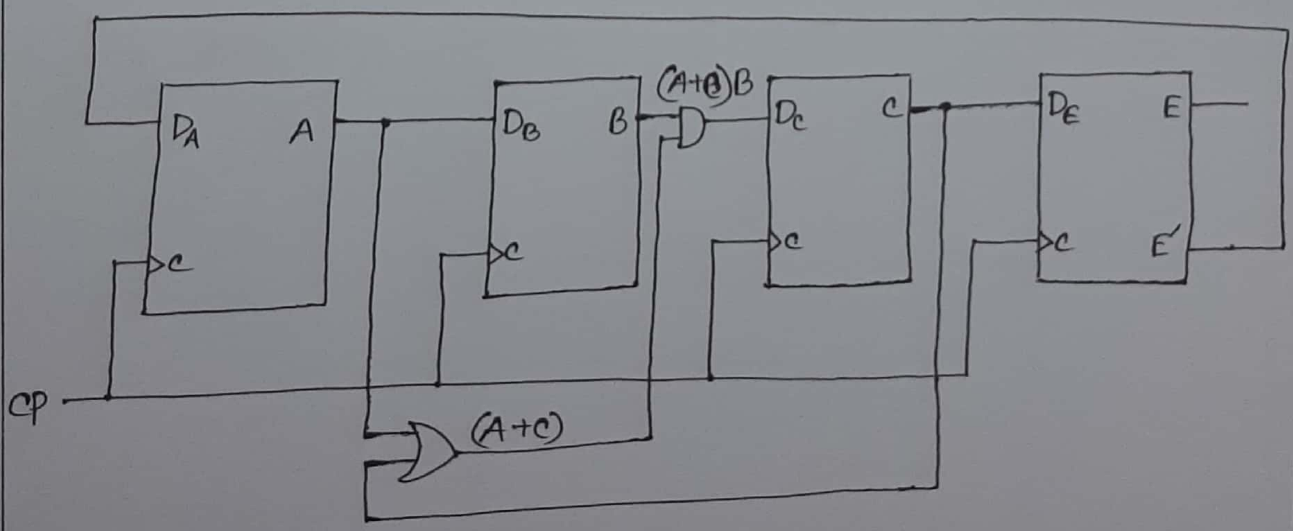


Fig: Modified Circuit for self correction

State Table for unused states:

Present state A B C E	Next State A(t+1) B(t+1) C(t+1) E(t+1)				Invalid States	Next state after modification
2 → 0 0 1 0 →	1	0	0	1 →	9	→ 1 0 0 1 (9)
4 → 0 1 0 0 →	1	0	1	0 →	10	→ 1 0 0 0 (8)
5 → 0 1 0 1 →	0	0	1	0 →	2	→ 0 0 0 0 (0)
6 → 0 1 1 0 →	1	0	1	1 →	11	→ 1 0 1 1 (11)
9 → 1 0 0 1 →	0	1	0	0 →	4	→ 0 1 0 0 (4)
10 → 1 0 1 0 →	1	1	0	1 →	13	→ 1 1 0 1 (13)
11 → 1 0 1 1 →	0	1	0	1 →	5	→ 0 1 0 1 (5)
13 → 1 1 0 1 →	0	1	1	0 →	6	→ 0 1 1 0 (6)

So, the correction of invalid/ unused states will occur in following sequence: 2 → 9 → 4 → 8

10 → 13 → 6 → 11 → 5 → 0

7.28: Construct a Johnson counter with ten timing signals.

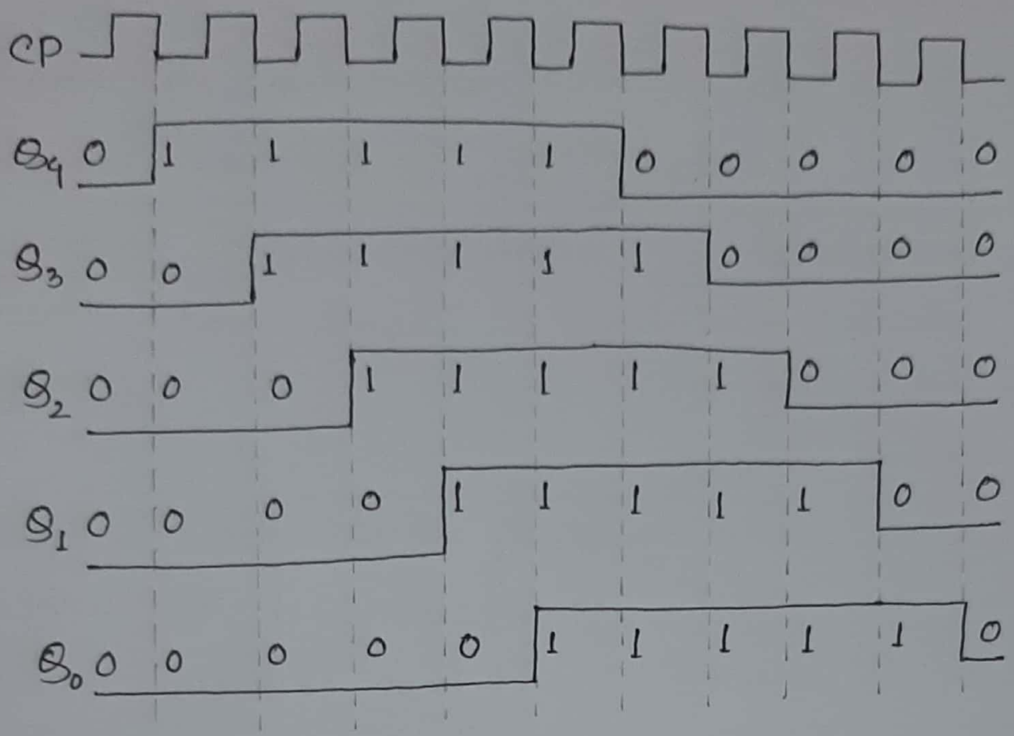
Answer: Johnson counters can be constructed for any number of timing sequence. The number of timing signals. So, we need five flip-flops to construct a Johnson counter with ten timing signals.

Count sequence:

Q_4	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	0
1	1	0	0	0
1	1	1	0	0
1	1	1	1	0
1	1	1	1	1
0	1	1	1	1
0	0	1	1	1
0	0	0	1	1
0	0	0	0	1

If we again complement the last bit of the number which is 1, then we will get 0 and hence the five bit number will become 00000 which is the initial value.

Diagram for timing signals:



Construction of the Johnson ~~circ~~ counter:

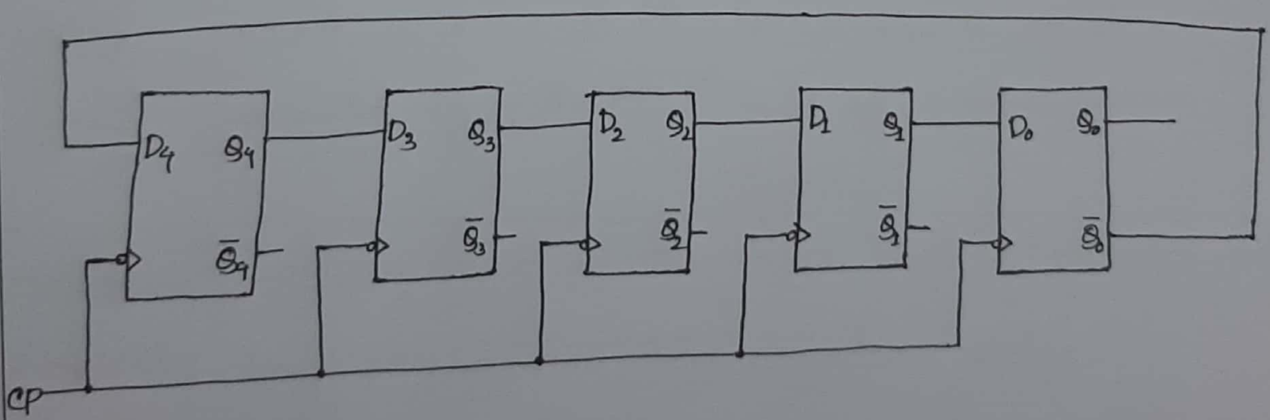


Fig: Johnson counter with 10 timing signals

7.33: Differentiate between Ripple counter and Synchronous counter.

Answer: The difference between Ripple counter and Synchronous counter is given below:

Parameter	Ripple Counter	Synchronous Counter
Circuit Complexity	Logic circuit is simple	With increase in number of states, the logic circuit becomes complicated.
Connection Pattern	Output of the preceding FF, is connected to clock of next FF.	There is no connection between output of preceding FF and CLK of next one.
Clock Input	All the FFs are not clocked simultaneously.	All FFs receive clock signal simultaneously.
Propagation Delay	P.D. = $n * (td)$ where, n is no. of FF and td is p.d. per FF.	P.D. = $n * (td)FF + (td)gate$ It is much shorter than that of asynchronous counter.
Maximum frequency of operation.	Low, because of the long propagation delay	High, due to shorter propagation delay.

7.34: Write short notes on-

a) Registers

Answer: A register is a quickly accessible location available to a computer's processor. Registers usually consist of a small amount of fast storage, although some registers have specific hardware functions, and may be read-only or write-only.

Registers are normally measured by the number of bits they can hold, for example, an "8-bit register", "32-bit register" or a "64-bit register" or even more. In some Instruction sets, the registers can operate in various modes breaking down its storage memory into smaller ones, 32-bit into four 8-bit ones for instance, to which multiple data can be loaded and operated upon at the same time. Typically it is implemented by adding extra registers that map their memory into bigger one. The number of registers available on a processor and the operations that can be performed using those registers has a significant impact on the efficiency of code generated by optimizing compilers.

b) Johnson counter

Answer: A Johnson counter is a modified is a ring counter in which the output from the last flip-flop is inverted and fed back as an input to the first. It is also called as Inverse feedback counter or Twisted Ring counter.

The Johnson counter circulates a stream of ones followed by zeros around the ring. For example, in a four register counter, with initial register values of 0000, the repeating pattern is- 0000, 1000, 1100, 1110, 1111, 0111, 0011, 0001, 0000...

Ring counters offer some advantages over binary counters; these require an adder circuit, which is substantially more complex. Aditionally, the worst-case propagation delay on an adder circuit is proportional to the number of bits in the code. The ring counter propagation delay is constant regardless of the number of bits in the code. The complex combinational logic of an adder can create timing errors which may cause erratic hardware performance. The disadvantage of ring counters is that they are lower density mo codes.

c) Ring counter

Answer: A ring counter is a type of counter composed of flip-flops connected into a shift register, with the output of the last flip-flop fed to the input of the first. making a "circular" or "ring" structure.

There are two types of ring counters. These are - straight ring counter (also known as a one-hot counter) and a twisted ring counter (also known as Johnson counter).

Ring counters are often used in hardware design to create finite-state machines. A binary counter would require an adder circuit which is substantially more complex than a ring counter and has higher propagation delay of a ring counter will be nearly constant regardless of the number of bits in the code.

A general disadvantage of ring counters is that they are lower density codes than normal binary encoding of state numbers.

Ring counter is a typical application of shift register. Sometimes bidirectional shift registers are used, so that bidirectional or updown ring counters can be made.

d) Random access memory

Answer: Random access memory (RAM) is a form of computer memory that can be read and changed in any order, typically used to store working data and machine code. A random access memory device allows data items to be read or written in almost the same amount of time irrespective of the physical location of data inside the memory, in contrast with other direct-access data storage media, where the time required to read and write data items varies significantly depending on their physical locations on the recording medium, due to mechanical limitations such as media rotation speeds and arm movement.

RAM contains multiplexing and demultiplexing circuitry, to connect the data lines to the addressed storage for reading or writing the entry. Usually more than one bit of storage is accessed by the same addess and RAM devices often have multiple data lines and are said to be "8-bit" or "16-bit" etc. devices.

In today's technology, random access memory takes the form of integrated circuit (IC) chips with MOS (metal-oxide-semiconductor) memory cells.

e) Parallel in Serial out (PISO) Shift Register

Answer: The parallel in/ serial out shift registers store data, shifts it on a clock by clock basis, and delays it by the number of stages times the clock period. In addition, parallel in/ serial out really means that we can load data in parallel into all stages before any shifting ever begins.

This way to convert data from a parallel format to a serial format. By parallel format we mean that the data bits are present simultaneously on individual wires, one for each data bit. By serial format we mean that the data bits are presented sequentially in time on a single wire or circuit as in the case of the "data out" on the block diagram below:

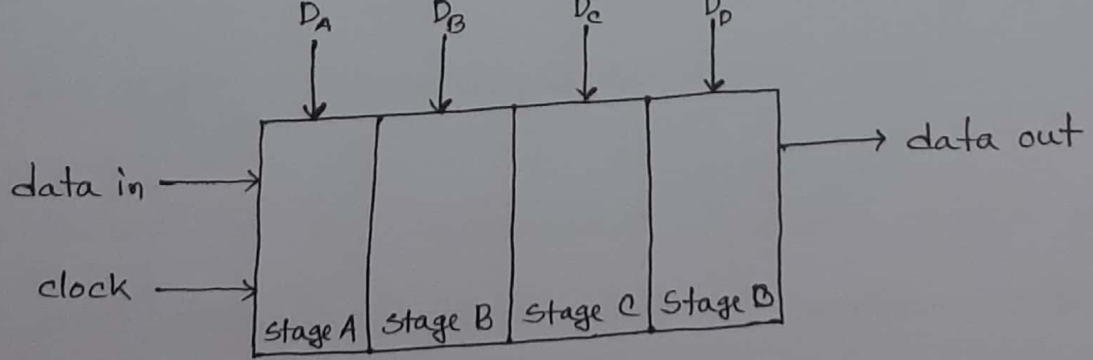
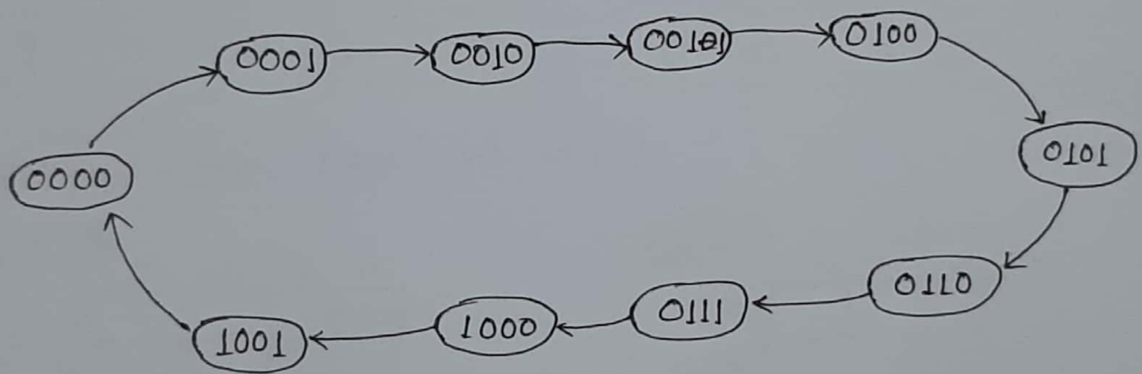


Fig: Parallel in Serial out shift register with 4-stages

7.38: Design a MOD 10 synchronous binary up counter using T flip-flop and other necessary logic gates. Draw the timing diagram.

Answer: There will be 10 states in a MOD 10 synchronous binary up counter. So, it will count from 0 to 9. Since it has 10 states, we will need 4 flip-flops to design the counter.

The state diagram:



Present State Q_n	Next State Q_{n+1}	Required Input T
0	0	0
0	1	1
1	0	1
1	1	0

State Table:

Present State				Next State				Required Input			
Q_4	Q_3	Q_2	Q_1	Q_4	Q_3	Q_2	Q_1	T_4	T_3	T_2	T_1
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	1
0	1	0	1	0	1	1	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	1
0	1	1	1	~~0~~ 1	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	1	0	0	0
1	0	0	1	0	0	0	0	1	0	0	1

K-maps for required input:

$Q_4Q_3 \backslash Q_2Q_1$	00	01	11	10
00				
01			1	
11	X	X	X	X
10	1	1	X	X

$T_4 = Q_4 + Q_3Q_2Q_1$

$Q_4Q_3 \backslash Q_2Q_1$	00	01	11	10
00			1	
01			1	
11	X	X	X	X
10		X	X	X

$T_3 = Q_2Q_1$

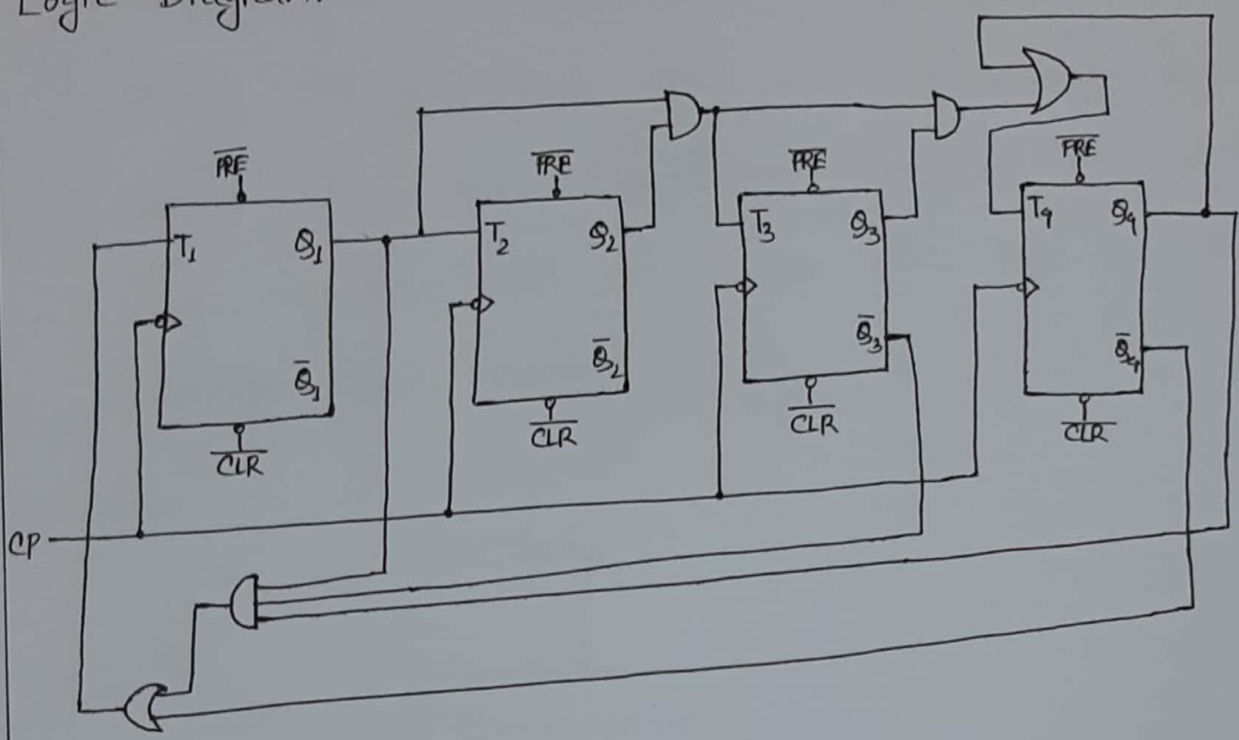
$Q_4Q_3 \backslash Q_2Q_1$	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	X	X	X	X
10		1	X	X

$T_1 = \bar{Q}_4 + Q_4\bar{Q}_3Q_1$

$Q_4Q_3 \backslash Q_2Q_1$	00	01	11	10
00		1	1	
01		1	1	
11	X	X	X	X
10		X	X	X

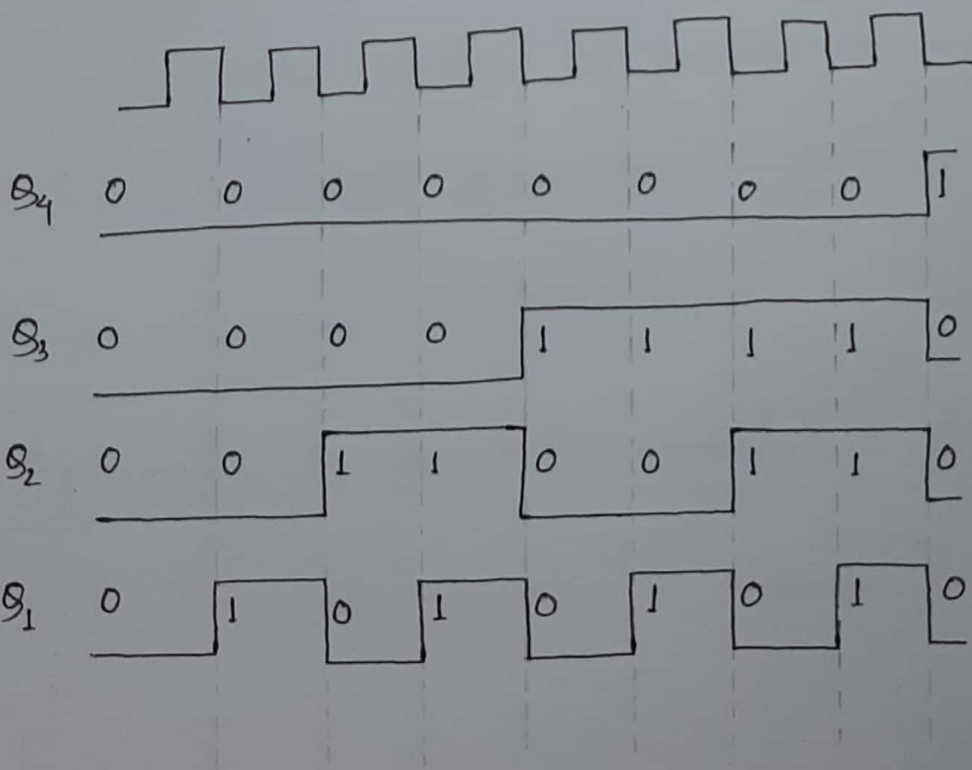
$T_2 = Q_1$

Logic Diagram:



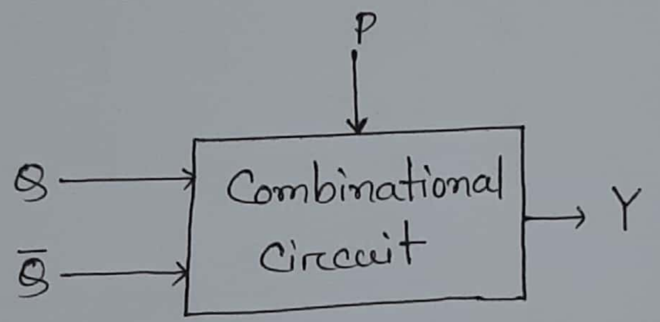
Timing Diagram:

Since there are 4 ffs, so there will be 8 timing signals



7.39: Design a 4-bit up/down a synchronous counter using JK flip-flops and other necessary logic gates. Use one directional control input. If P=0, the counter will count up and for P=1, the counter will count down.

Answer:



here,

Q, $\bar{Q}$ are the output of previous ff

Y is the output given to the next ff

P is a mode control input.

If, P = 0 ⟶ Up count (Q connected to CP)

P = 1 ⟶ Down count ($\bar{Q}$ connected to CP)

MP	Q	$\bar{Q}$	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

P \ $Q\bar{Q}$				
	0	0	1	1
	0	1	1	0

$$Y = \bar{M}Q + M\bar{Q}$$

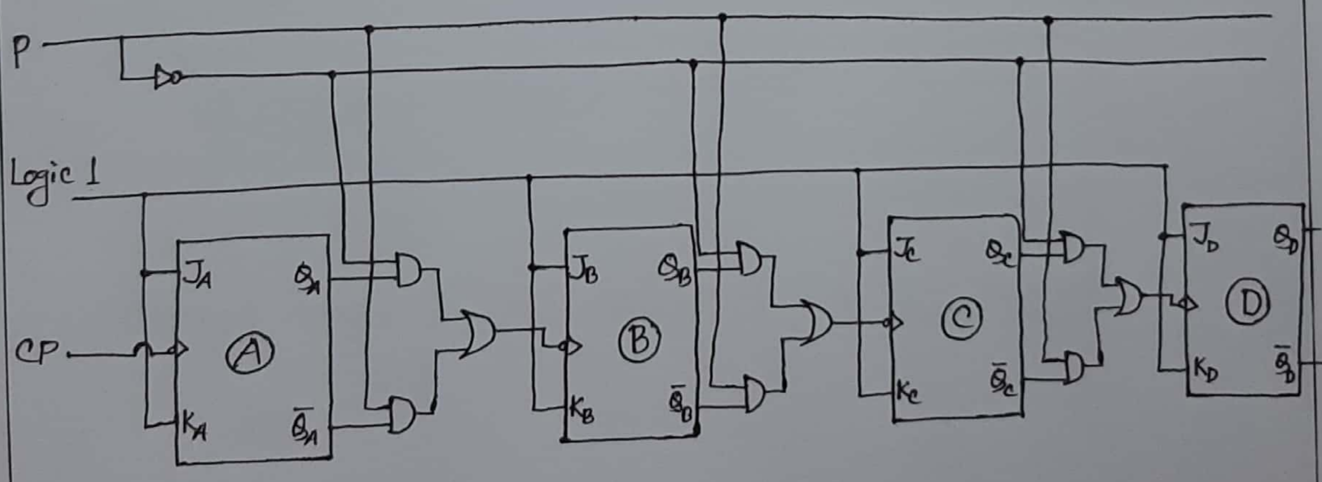


Fig: 4-bit up/down asynchronous counter using JK flip-flops.